Fundamental Group of the Torus via Seifert–van Kampen

An End-to-End Meta-Math Walkthrough

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(Dated: October 2026)

Abstract

A complete authored walkthrough of fundamental group of the torus via seifert–van kampen, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For the torus T^2, prove*

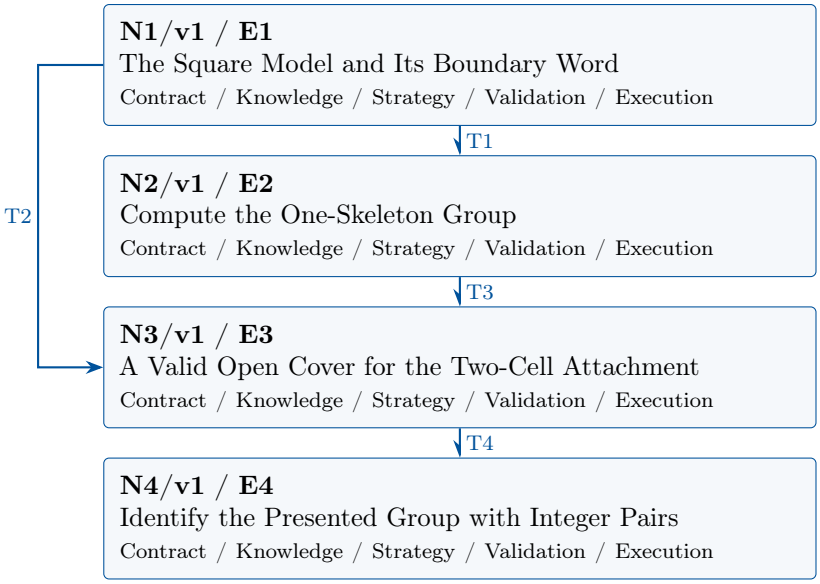
$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z}^2.$$

The output is a fundamental-group presentation followed by an explicit group isomorphism. A drawing of two loops or the assertion that they commute is not a proof that these are all generators and relations.

A. Foundations and permitted imports

a based loop is considered modulo based homotopy; multiplication is loop concatenation. Import $\pi_1(S^1) \cong \mathbb{Z}$, homotopy invariance, and Seifert–van Kampen for path-connected open sets with path-connected intersection. For such a cover, the theorem gives the group pushout: relations identify the two images of the intersection group. We explicitly construct the cover used for the torus's two-cell attachment.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The torus has one vertex, loops a, b, and one two-cell attached by $aba^{-1}b^{-1}$.

Receiving activity. N2: Compute the One-Skeleton Group

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, the circle group, and Seifert–van Kampen.

T2: N1/E1/v1 to N3. The torus has one vertex, loops a, b, and one two-cell attached by $aba^{-1}b^{-1}$.

Receiving activity. N3: A Valid Open Cover for the Two-Cell Attachment

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, and van Kampen with a valid open cover.

T3: N2/E2/v1 to N3. The one-skeleton has fundamental group $F(a, b)$.

Receiving activity. N3: A Valid Open Cover for the Two-Cell Attachment

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, and van Kampen with a valid open cover.

T4: N3/E3/v1 to N4. $\pi_1(T^2) = F(a, b)/\langle\langle aba^{-1}b^{-1}\rangle\rangle$.

Receiving activity. N4: Identify the Presented Group with Integer Pairs

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N3/E3 and the universal property of a group presentation.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: The Square Model and Its Boundary Word

Object and role. The Square Model and Its Boundary Word. Establish the following local result: The torus has one vertex, loops a, b, and one two-cell attached by $aba^{-1}b^{-1}$.

Inputs. The quotient square with opposite edges consistently identified.

C — Contract

Own E1/v1: The torus has one vertex, loops a, b, and one two-cell attached by $aba^{-1}b^{-1}$.

K — Knowledge

Quotient square and orientations

S — Strategy

Separate one-skeleton and face

V — Validation

Identify all four vertices

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Track all four vertices to the common vertex.

N1.2 Choose positive horizontal and vertical loop orientations.

N1.3 Read the counterclockwise boundary word with its inverse edges.

Export E1/v1. The torus has one vertex, loops a, b, and one two-cell attached by $aba^{-1}b^{-1}$.

Acceptance gate. Identify all four vertices Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Identify all four vertices Recheck dependent nodes N2, N3, N4.

B. N2/v1: Compute the One-Skeleton Group

Object and role. Compute the One-Skeleton Group. Establish the following local result: The one-skeleton has fundamental group $F(a, b)$.

Inputs. N1/E1, the circle group, and Seifert–van Kampen.

C — Contract

Own E2/v1: The one-skeleton has fundamental group $F(a, b)$.

K — Knowledge

Circle group and van Kampen

S — Strategy

Use wedge-of-circles cover

V — Validation

Do not assume a and b commute

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Use two open sets retracting to the individual circles.

N2.2 Check their intersection is a contractible tree.

N2.3 The pushout is the free product $\mathbb{Z} * \mathbb{Z}$, before imposing a face relation.

Export E2/v1. The one-skeleton has fundamental group $F(a, b)$.

Acceptance gate. Do not assume a and b commute Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Do not assume a and b commute Recheck dependent nodes N3, N4.

C. N3/v1: A Valid Open Cover for the Two-Cell Attachment

Object and role. A Valid Open Cover for the Two-Cell Attachment. Establish the following local result: $\pi_1(T^2) = F(a,b)/\langle\langle aba^{-1}b^{-1}\rangle\rangle$.

Inputs. N1/E1, N2/E2, and van Kampen with a valid open cover.

C — Contract

Own E3/v1: $\pi_1(T^2) = F(a,b)/\langle\langle aba^{-1}b^{-1}\rangle\rangle$.

K — Knowledge

Open-cover pushout

S — Strategy

Fill the two-cell

V — Validation

Check open sets and connected intersection

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Take the complement of the face center and a small open disk around that center.

N3.2 The punctured overlap is path connected; its generator maps to the attaching word and to the identity respectively.

N3.3 Form the normal closure relation, including the basepoint conjugation.

Export E3/v1. $\pi_1(T^2) = F(a,b)/\langle\langle aba^{-1}b^{-1}\rangle\rangle$.

Acceptance gate. Check open sets and connected intersection Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Check open sets and connected intersection Recheck dependent nodes N4.

D. N4/v1: Identify the Presented Group with Integer Pairs

Object and role. Identify the Presented Group with Integer Pairs. Establish the following local result: $\pi_1(T^2) \cong \mathbb{Z}^2$.

Inputs. N3/E3 and the universal property of a group presentation.

C — Contract

Own E4/v1: $\pi_1(T^2) \cong \mathbb{Z}^2$.

K — Knowledge

Presentations and integer pairs

S — Strategy

Construct inverse group maps

V — Validation

Prove no additional hidden relations

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Map a, b to the standard integer coordinate vectors.

N4.2 Define the reverse map $(m, n) \mapsto a^m b^n$, using commutativity.

N4.3 Check both composites are identities, ruling out further integer relations.

Export E4/v1. $\pi_1(T^2) \cong \mathbb{Z}^2$.

Acceptance gate. Prove no additional hidden relations Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Prove no additional hidden relations Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — The Square Model and Its Boundary Word

Represent the torus by a square with opposite edges identified in the same direction. All four vertices become one vertex, and the horizontal and vertical edge pairs become loops a and b. The one-skeleton Y is the wedge $S^1 \vee S^1$.

The remaining open square is one two-cell. Traversing its boundary counterclockwise gives horizontal a, vertical b, reverse horizontal $a^{(-1)}$, and reverse vertical $b^{(-1)}$. Hence its attaching loop is

$$w = aba^{-1}b^{-1}.$$

Changing consistent orientation conventions can invert or conjugate the word without changing its normal closure. Inconsistent conventions, however, can produce the wrong space.

B. N2 — Compute the One-Skeleton Group

Cover the wedge by one open set containing the entire first circle and short arcs around the wedge point on the second, and a second set with the roles reversed. Each deformation retracts onto its full circle. Their intersection is a small cross-shaped tree around the common vertex, which is contractible and path connected.

Van Kampen therefore gives

$$\pi_1(Y) \cong \mathbb{Z} * \mathbb{Z} = F(a, b),$$

the free group on a and b. Before the face is filled, the generators do not commute; the commutator is a nontrivial reduced word in this free group.

C. N3 — A Valid Open Cover for the Two-Cell Attachment

Write X for Y with the closed disk attached along w; this is the square model of the torus. Choose an interior center o of the attached disk.

Let $U = X \setminus \{o\}$. This is open and deformation retracts onto Y: radially push each point in the punctured disk toward its boundary while fixing Y. Let V be a smaller open disk around o lying wholly inside the two-cell. It is open and contractible. These sets cover X. Their intersection is a punctured smaller disk, path connected, with fundamental group Z.

Choose a basepoint in that intersection and a path in U to the original vertex. A positive generator of the intersection group, transported to Y by the radial retraction, becomes w up to this basepoint conjugation. Its image in V is trivial because V is contractible.

Van Kampen identifies the U-image with the trivial V-image. Consequently

$$\pi_1(X) \cong F(a, b)/\langle\langle w \rangle\rangle,$$

where the double brackets mean the normal subgroup generated by w. Normal closure is required because the fundamental group relation also kills all conjugates of the attaching loop. This proves the stated presentation and shows that the single two-cell introduces exactly that normal relation.

D. N4 — Identify the Presented Group with Integer Pairs

Let $P = \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle$. The relation is equivalent to $ab = ba$. Define

$$\Phi : P \to \mathbb{Z}^2, \qquad \Phi(a) = (1, 0), \quad \Phi(b) = (0, 1).$$

It is well-defined because the commutator maps to zero in the abelian target. Define the opposite map

$$\Psi : \mathbb{Z}^2 \to P, \qquad \Psi(m, n) = a^m b^n.$$

Since a and b commute, Psi respects addition: $a^m b^n a^r b^s = a^{m+r} b^{n+s}$. Phi composed with Psi is the identity on integer pairs. Psi composed with Phi is the identity on the generators a,b and hence on all of P. Thus the maps are inverse isomorphisms. No extra relation between the integers remains.

Combining N1–N4 proves $\pi_1(T^2) \cong \mathbb{Z}^2$. □

□

V. WORKED ORCHESTRATION

In the product model $T^2 = S^1 \times S^1$ with basepoint (1,1), define

$$\gamma_{m,n}(t) = (e^{2\pi i m t}, e^{2\pi i n t}), \qquad 0 \le t \le 1.$$

Its class corresponds to (m,n). The coordinate projection maps send a to winding number 1 in the first circle and b to winding number 1 in the second, so they realize the same Phi constructed in N4.

- **Loop word**

  a

  Integer pair

  (1,0)

  Consequence

  Horizontal generator

- **Loop word**

  b

Integer pair

(0,1)

Consequence

Vertical generator

- **Loop word**

  $a^2b^{(-1)}$

  Integer pair

  $(2, -1)$

  Consequence

  Not null-homotopic

- **Loop word**

  $aba^{(-1)}b^{(-1)}$

  Integer pair

  (0,0)

  Consequence

  Killed by the attached face

- **Loop word**

  $a^2b^{(-1)}$ followed by $a^{(-1)}b^3$

  Integer pair

  (1,2)

  Consequence

  Concatenation becomes addition

Because Phi is an isomorphism, a loop is null-homotopic exactly when both integer coordinates vanish. The zero-pair conclusion is supported by injectivity, not merely by observing two zero winding numbers.

A. Boundary and hypothesis audit

A cover with disconnected intersection does not satisfy the stated version of van Kampen. Reading the face word as ab instead of the commutator changes the presentation and yields a different calculation. Homology abelianization alone cannot recover an arbitrary fundamental group; here the presentation proves that the full group itself is abelian.

Export the square cell model, free one-skeleton group, explicit attaching-relation calculation, and isomorphism with Z^2. The same cell structure supports the torus homology problem, where signed incidences give a different but compatible invariant.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

A failed branch abelianizes $F(a, b)$ before accounting for the face, claiming that any two-loop graph has group Z^2.

- **Node:** N2
- **State:** STALLED
- **Failing component:** Execution
- **Failed claim:** wedge group is Z squared
- **Observed:** commutator is nontrivial reduced word before face attachment
- **Repair target:** cell-attachment dependency
- **Blocked export:** torus group without filling argument

Knowledge retrieves the free-product conclusion of van Kampen. Validation distinguishes the one-skeleton from the whole torus. Strategy restores the face-attachment branch. Execution applies the valid open cover, imposes the normal commutator relation, and checks both inverse maps. Contract publishes Z^2 only after this dependency is satisfied.

B. Hypothesis boundary

A cover with disconnected intersection does not satisfy the stated version of van Kampen. Reading the face word as ab instead of the commutator changes the presentation and yields a different calculation. Homology abelianization alone cannot recover an arbitrary fundamental group; here the presentation proves that the full group itself is abelian.

Export the square cell model, free one-skeleton group, explicit attaching-relation calculation, and isomorphism with Z^2. The same cell structure supports the torus homology problem, where signed incidences give a different but compatible invariant.

C. Version and recheck

Version changed contracts and recheck the revised export and every dependent node. Diagnostic feedback is not a proof premise.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Allen Hatcher, Algebraic Topology, Chapter 1: van Kampen and cell attachments.